

Deploying an ML Model on AWS

The two main things I would consider first is the Sentiment Analysis's model size and memory requirement. Model size would affect how long it takes to load the model and storage requirement. Meanwhile, max memory requirement for inference would need to be analysed first depending on what the intended use case are for, whether it will be for single inference or batch inference.

Assuming a model size of <10GB and memory consumption <10GB, I would use an image based lambda. The model would be stored in a Docker container hosted in ECR, which Lambda would run as its environment. Using a container would provide precise control of the environment and also better cold start compared to the overhead if the model were to be downloaded from S3 everytime. This lambda function can then be invoked through API Gateway. A simple .parquet log could also be stored in S3 and then made queryable through Athena.

If the model requires GPU support or requires longer processing time than Lambda's limit, only then would I propose to use Sagemaker as cost of keeping an instance running is charged by the hour.